

## **WEEK4: Basics of Django Framework & Installation of required Software's**

### **Step1: install Python (latest version)**

python --version

pip --version

pip install virtualenv or venv

### **Create parent directory to save all Django projects**

mkdir my\_django\_project

cd my\_django\_project

### **Step2: create virtual Environment**

C:\>python -m venv week4

### **Next, Activate the environment**

.\env\Scripts\activate

Or

C:\> cd week4

C:\week4\>cd Scripts

C:\> week4\Scripts\.\activate.bat or .ps1(powershell)

### **Step3 : install django**

pip install django

### **to know the version**

django-admin --version

### **step4:create Django project**

django-admin startproject website ← project name

cd website

to check successfully installed Django or not runserver

**python manage.py runserver**

then get Django welcome page – means successfully django installed

step4: create simple application -

#### **step4.1 To Create app**

#to create app –

python manage.py startapp myapp

*Create these files separately*

*HTML- index.html*

*CSS-style.css*

*Javascript- script.js*

**Step 4.2 create subfolders in myapp**

**static – add CSS and js files**

**templates –HTML files**

#### **Project Structure**

website/

├─ myapp/

| └─ templates/

| | └─ myapp/

| | └─ index.html

```
|  └─ static/
|  |   └─ myapp/
|  |       |   └─ style.css
|  |       |   └─ script.js
```

In Templates\myapp\Index.html

```
{% load static %}
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="{% static 'myapp/style.css' %}">
  <script src="{% static 'myapp/script.js' %}"></script>
</head>
<body>
  <h1>Welcome to My Django App</h1>
  <button onclick="showMessage()">Click Me</button>
  <p id="output"></p>
</body>
</html>
```

In Static\myapp\script.js

```
function showMessage() {
  document.getElementById("output").innerText = "Hello from JavaScript!";
}
```

In Static\myapp\style.css

```
body {
  font-family: Arial;
  background-color: #f0f0f0;
}
button {
  padding: 10px;
  background-color: #007bff;
  color: white;
  border: none;
}
```

## Django Templates

Templates are HTML files with Django's own template language (DTL) embedded. They allow you to:

- Insert dynamic data using `{{ variable }}`
- Control logic with `{% tags %}` like loops and conditionals
- Extend base layouts using `{% extends %}` and `{% block %}`

### Step5 :Configure Template directory in settings.py

```
TEMPLATES = [  
  
    {  
  
        'BACKEND': 'django.template.backends.django.DjangoTemplates',  
  
        'DIRS': [BASE_DIR / 'templates'], # Your custom templates folder  
  
        'APP_DIRS': True,  
  
        'OPTIONS': {  
  
            'context_processors': [  
  
                'django.template.context_processors.debug',  
  
                'django.template.context_processors.request',  
  
                'django.contrib.auth.context_processors.auth',  
  
                'django.contrib.messages.context_processors.messages',  
  
            ],  
  
        },  
  
    ],  
]
```

### For Static directory configure in settings.py

```
STATIC_URL = '/static/'
```

```
STATICFILES_DIRS = [  
    BASE_DIR / 'static',  
]
```

**OR**

```
STATICFILES_DIRS = [  
    BASE_DIR / 'myapp' / 'static',  
]
```

### Step 6: Add myapp in installedApps settings.py

#Application definition

```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
    'myapp',  
]
```

**Step7: In Django, views.py and urls.py are essential for handling how your web app responds to user requests.**

### Views.py

This file contains **view functions or classes** that define what happens when a user visits a specific URL.

**Purpose:** It processes requests and returns responses.

- **Common tasks:**
- Fetch data from the database
- Render templates (HTML pages)
- Handle form submissions
- Return JSON for APIs

```
# views.py
from django.shortcuts import render

def index(request):
    return render(request, 'myapp/index.html')
```

## urls.py The Routing System

This file maps **URL patterns** to the corresponding view functions or classes.

- **Purpose:** It connects user requests (like /home) to the right logic in views.py.
- **Structure:** You define urlpatterns using Django's path() or re\_path().

### Example:

If not exists , we need to create urls.py in myapp

**Myapp/urls.py**- This file is **specific to one app** and defines the routes for that app's views

```
# urls.py

from django.urls import path
from . import views

urlpatterns = [
    path("", views.index, name='index'),
]
```

You can also include app-specific urls.py files in your project-level urls.py using include():

**website/urls.py** - This file is the **central URL dispatcher** for your entire Django project

```
# project-level urls.py

from django.contrib import admin
from django.urls import path, include

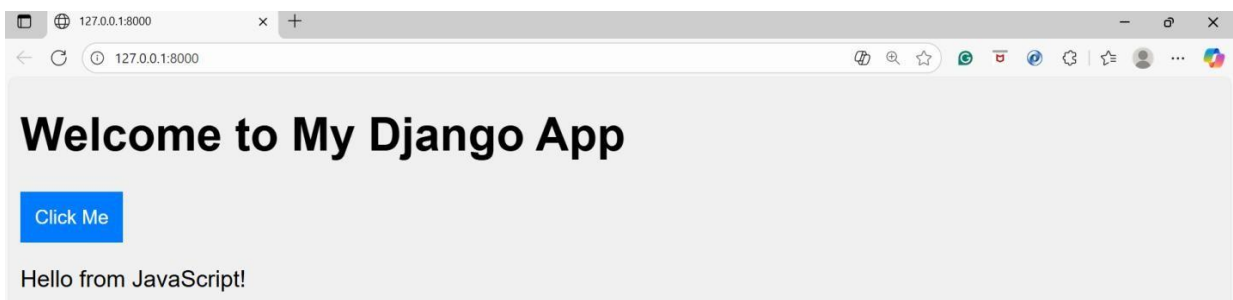
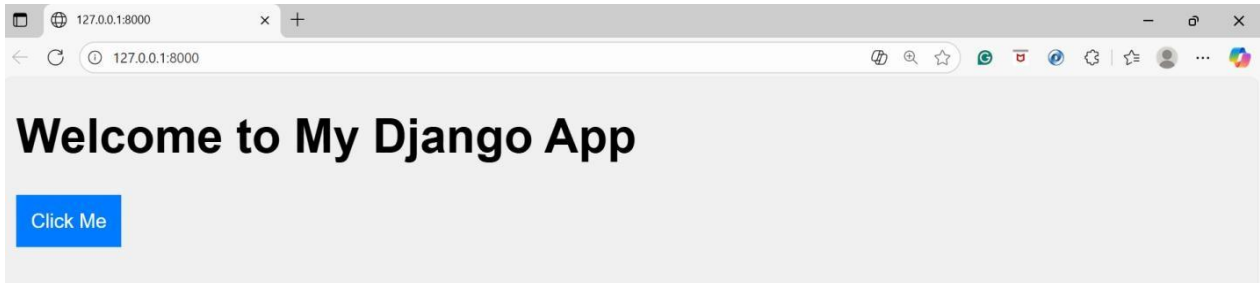
urlpatterns = [
    path('admin/', admin.site.urls),
```

```
    path('', include('myapp.urls')),  
]
```

### Step9: Run application/project

```
python manage.py runserver
```

### OUTPUT:



**WEEK5:** Create a shopping cart web application and add a customized admin page for product catalog management for Experiment 3

**Project Structure:**

shopping\_cart/

```
├── manage.py
├── shopping_cart/      # Project settings
│   ├── __init__.py
│   ├── settings.py
│   ├── urls.py
│   └── wsgi.py
└── shop/              # Main app
    ├── __init__.py
    ├── admin.py
    ├── apps.py
    ├── models.py
    ├── views.py
    ├── urls.py
    ├── forms.py
    └── templates/
        ├── base.html
        ├── registration.html
        ├── login.html
        ├── catalog.html
        └── cart.html
```



Step1:

```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
    'shop',  
]  
LOGIN_REDIRECT_URL = 'catalog'  
LOGOUT_REDIRECT_URL = 'login'
```

**Step2:**

```
#views.py  
from django.shortcuts import render, redirect, get_object_or_404  
from django.contrib.auth import login  
#from django.contrib.auth import logout  
from django.shortcuts import redirect  
from django.contrib.auth.decorators import login_required  
from .forms import RegistrationForm, CartItemForm  
from .models import Product, CartItem
```

```
def register(request):  
    if request.method == 'POST':  
        form = RegistrationForm(request.POST)  
        if form.is_valid():  
            user = form.save()  
            login(request, user)  
            return redirect('catalog')  
    else:  
        form = RegistrationForm()  
    return render(request, 'registration.html', {'form': form})
```

```
@login_required  
def catalog(request):  
    products = Product.objects.all()  
    return render(request, 'catalog.html', {'products': products})
```

```
@login_required  
def add_to_cart(request, product_id):
```

```

    product = get_object_or_404(Product, id=product_id)
    cart_item, created = CartItem.objects.get_or_create(user=request.user,
product=product)
    if not created:
        cart_item.quantity += 1
    cart_item.save()
    return redirect('cart')

```

```

@login_required
def cart(request):
    items = CartItem.objects.filter(user=request.user)
    total = sum(item.subtotal() for item in items)
    return render(request, 'cart.html', {'items': items, 'total': total})

```

```

@login_required
def update_cart(request, item_id):
    item = get_object_or_404(CartItem, id=item_id, user=request.user)
    if request.method == 'POST':
        form = CartItemForm(request.POST, instance=item)
        if form.is_valid():
            form.save()
    return redirect('cart')
#def user_logout(request):

```

```

    logout(request) # clears the session
    return redirect('catalog') # redirect to catalog or homepage

```

```

#models.py
from django.db import models
from django.contrib.auth.models import User

```

```

class Product(models.Model):
    name = models.CharField(max_length=100)
    price = models.DecimalField(max_digits=8, decimal_places=2)
    description = models.TextField(blank=True)

    def __str__(self):
        return self.name

```

```

class CartItem(models.Model):
    user = models.ForeignKey(User, on_delete=models.CASCADE)
    product = models.ForeignKey(Product, on_delete=models.CASCADE)
    quantity = models.PositiveIntegerField(default=1)

```

```

    def subtotal(self):
        return self.product.price * self.quantity

#forms.py
from django import forms
from django.contrib.auth.models import User
from django.contrib.auth.forms import UserCreationForm
from .models import CartItem

class RegistrationForm(UserCreationForm):
    email = forms.EmailField(required=True)

    class Meta:
        model = User
        fields = ['username', 'email', 'password1', 'password2']

class CartItemForm(forms.ModelForm):
    class Meta:
        model = CartItem
        fields = ['quantity']

# admin.py
from django.contrib import admin
from .models import Product, CartItem

@admin.register(Product)
class ProductAdmin(admin.ModelAdmin):
    list_display = ('name', 'price', 'description')
    search_fields = ('name',)

@admin.register(CartItem)
class CartItemAdmin(admin.ModelAdmin):
    list_display = ('user', 'product', 'quantity', 'subtotal')

shop/urls.py
from django.urls import path
from django.contrib.auth import views as auth_views
from . import views

urlpatterns = [
    path('register/', views.register, name='register'),

```

```

    path('catalog/', views.catalog, name='catalog'),
    path('cart/', views.cart, name='cart'),
    path('add/<int:product_id>/', views.add_to_cart, name='add_to_cart'),
    path('update/<int:item_id>/', views.update_cart, name='update_cart'),
    path('login/', auth_views.LoginView.as_view(template_name='login.html'),
name='login'),
    path('logout/', auth_views.LogoutView.as_view(), name='logout'),

```

```

]
Shopping_cart/Urls.py
from django.contrib import admin
from django.urls import path, include

```

```

urlpatterns = [
    path('admin/', admin.site.urls),
    path("", include('shop.urls')),

```

]
**In shop/templates folder**

```

Base.html
<!DOCTYPE html>
<html>
<head>
    <title>Shopping Cart</title>
</head>
<body>
    <nav>
        <a href="{% url 'catalog' %}">Catalog</a> |
        <a href="{% url 'cart' %}">Cart</a> |
        {% if user.is_authenticated %}
            <a href="{% url 'logout' %}">Logout</a>
        {% else %}
            <a href="{% url 'login' %}">Login</a> |
            <a href="{% url 'register' %}">Register</a>
        {% endif %}
    </nav>
    <hr>
    {% block content %}{% endblock %}
</body>
</html>

```

Registration.html

```

{% extends 'base.html' %}
{% block content %}

```

```

<h2>Create an Account</h2>
<form method="post">
  {% csrf_token %}
  {{ form.as_p }}
  <button type="submit">Register</button>
</form>
<p>Already have an account? <a href="{% url 'login' %}">Login here</a>.</p>
{% endblock %}

```

### Login.html

```

{% extends 'base.html' %}
{% block content %}
<h2>Login</h2>
<form method="post">
  {% csrf_token %}
  {{ form.as_p }}
  <button type="submit">Login</button>
</form>
<p>Don't have an account yet? <a href="{% url 'register' %}">Register
here</a>.</p>
{% endblock %}

```

### Catalog.html

```

{% extends 'base.html' %}
{% block content %}
<h2>Product Catalog</h2>
<ul>
  {% for product in products %}
    <li>{{ product.name }} - ${{ product.price }}
      <a href="{% url 'add_to_cart' product.id %}">Add to Cart</a>
    </li>
  {% endfor %}
</ul>
{% endblock %}

```

### Cart.html

```

{% extends 'base.html' %}
{% block content %}
<h2>Your Cart</h2>
<ul>
  {% for item in items %}
    <li>
      {{ item.product.name }} (x{{ item.quantity }}) - ${{ item.subtotal }}
      <form method="post" action="{% url 'update_cart' item.id %}">
        {% csrf_token %}

```

```

        {{ item.quantity }}
        <input type="number" name="quantity" value="{{ item.quantity }}"
min="1">
        <button type="submit">Update</button>
    </form>
</li>
{% endfor %}
</ul>
<p>Total: ${{ total }}</p>
{% endblock %}

```

### Commands

python manage.py makemigrations

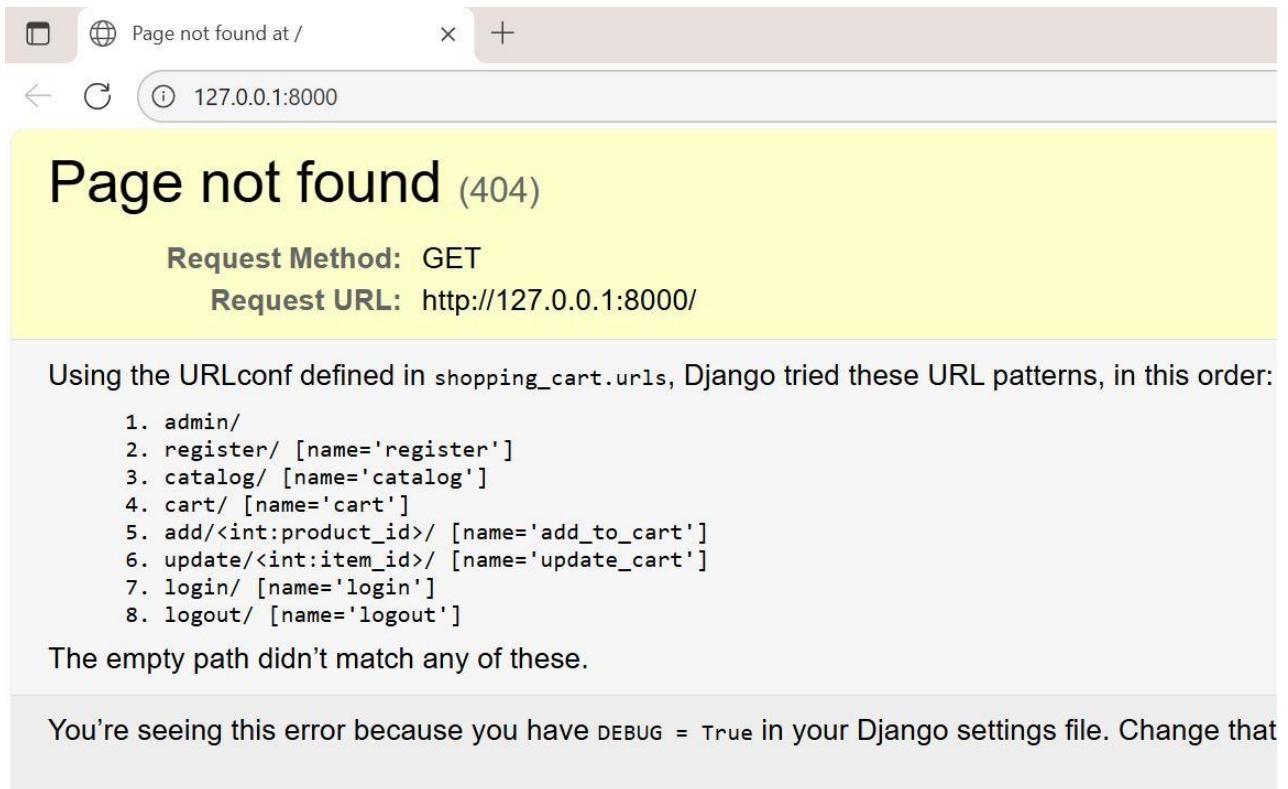
python manage.py migrate

### Create Superuser:

python manage.py createsuperuser

### Run Application:

python manage.py runserver



Shop administration | Django site

127.0.0.1:8000/admin/shop/

Sun

Django administration

Home > Shop

Shop administration

SHOP

Cart items

+ Add

Change

Products

+ Add

Change

Add product | Django site admin

127.0.0.1:8000/admin/shop/product/add/

Summarize

Copilot

Django administration

WELCOME, 91944. VIEW SITE / CHANGE PASSWORD / LOG OUT

Home > Shop > Products > Add product

Start typing to filter...

AUTHENTICATION AND AUTHORIZATION

Groups + Add

Users + Add

SHOP

Cart items + Add

Products + Add

Add product

Name:

Price:

Description:

SAVE

Save and add another

Save and continue editing

Select product to change | Django

127.0.0.1:8000/admin/shop/product/

Summarize

🔍

☆

🌐

🔒

⚙️

🔖

👤

⋮

🌈

Co

Django administration

WELCOME, 91944 / VIEW SITE / CHANGE PASSWORD / LOG OUT

Home > Shop > Products

Start typing to filter...

AUTHENTICATION AND AUTHORIZATION

Groups + Add

Users + Add

SHOP

Cart items + Add

Products + Add

✔️ The product "Head Phones" was added successfully.

Select product to change

ADD PRODUCT +

🔍

Search

Action: 

-----

Go

 0 of 3 selected

| <input type="checkbox"/> | NAME        | PRICE    | DESCRIPTION           |
|--------------------------|-------------|----------|-----------------------|
| <input type="checkbox"/> | Head Phones | 2000.00  | JBL                   |
| <input type="checkbox"/> | Mobile      | 28000.00 | SAMSUNG M35           |
| <input type="checkbox"/> | laptop      | 56000.00 | I7 , SSD 256GB HP 14" |

3 products



[Catalog](#) | [Cart](#) | [Logout](#)

## Product Catalog

- laptop - \$56000.00 [Add to Cart](#)
- Mobile - \$28000.00 [Add to Cart](#)
- Head Phones - \$2000.00 [Add to Cart](#)

[Catalog](#) | [Cart](#) | [Logout](#)

## Your Cart

- Head Phones (x1) - \$2000.00

1  Update

Total: \$2000.00

Page not found at /update/3 add more items to cart redirected to cart page automatically

 Shopping Cart × +

← ↺ (i) 127.0.0.1:8000/cart/

[Catalog](#) | [Cart](#) | [Logout](#)

---

## Your Cart

- Head Phones (x3) - \$6000.00  
3  Update

Total: \$6000.00

